Ashrith Edukulla

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Education

University of Michigan, Ann Arbor *Aug 2024 – Dec 2026*

B.E. in Robotics | GPA: 3.8/4.0

- Honors: Rank 3 (WRO), 4× IYRC medalist, 3× AMC qualifier, 2× AIME qualifier.

Work Experience

Field Robotics Group (FRoG) Lab @ University of Michigan *Dec 2025 – Present*

Robotics Software Engineer – Computer Vision & Active Perception

- Designing a **multimodal dual-map framework** fusing sonar and stereo vision for underwater submerged object inspection, achieving **40% improvement** in surface coverage over single-modality sonar-only baselines in simulation.
- Developing **frontier-based active perception** with occupancy grid updates and uncertainty-aware coverage metrics, reducing redundant re-exploration by **35%** across cluttered low-visibility underwater inspection environments.
- Implementing **3D SLAM** via SLIM-VDB with boustrophedon decomposition for real-time coverage planning, maintaining consistent probabilistic map estimates at scale across **200m^2** simulated inspection zones.
- Building a **dual stereo-sonar perception pipeline** dynamically transitioning between sonar-driven global exploration and stereo vision-guided close-range inspection passes, cutting total mission time by **28%**.
- Benchmarking full autonomy stack in **OceanSim** (Nvidia Isaac Sim) against single-map baselines across 10+ scenarios with varying turbidity, structural occlusion, and calibrated multi-axis sensor noise perturbations.

EECS 467: Autonomous Robotics @ University of Michigan *Dec 2025 – May 2026*

Instructional Aide

- Assisted instruction in **autonomous robotics**, covering perception, SLAM, motion planning, and feedback systems.
- Supported students in **ROS-based labs**, debugging, and end-to-end algorithm implementation on mobile robots.
- Held weekly office hours guiding **multi-sensor fusion**, navigation pipelines, and robot software architecture design.

Synergic Adaptive Machinas (SAM) Lab @ University of Michigan *Aug 2025 – Dec 2025*

Robotics Research Engineer

- Built a reconfigurable **3×3 mechatronic platform** to study precision motion, coordination, and system dynamics.
- Developed decentralized **C++ firmware** for 9-motor actuation with calibrated PWM control, and timing.
- Created a **C++/Sockets/Serial** interface to command actuators, log data, and support experiment pipelines.
- Analyzed distributed-control strategies, interaction effects, and **motion profiles** for stable closed-loop behavior.
- Co-authoring a paper on coordinated motion, **energy-aware control**, and scalable multi-actuator architectures.

ROB 201: Robot Differential Equations @ University of Michigan *Aug 2025 – Dec 2025*

Instructional Aide

- Supported instruction in **ODE modeling**, state-space analysis, and core control foundations for dynamic systems.
- Led weekly office hours using **Julia** simulations, visualizations, and interactive differential-equation problem solving.
- Assisted with labs and grading; clarified **Laplace transforms**, transfer functions, and fundamental PID design.

ROB 101: Computational Linear Algebra @ University of Michigan *Aug 2025 – Dec 2025*

Instructional Aide

- Supported instruction in **computational linear algebra**, matrices, vectors, and systems.
- Led weekly office hours focused on **Julia** coding, linear systems, regression, and data reasoning.
- Assisted with labs and grading; clarified **matrix operations**, optimization, and spatial coordinate concepts.

Michigan Mars ROVER *Nov 2024 – May 2025*

Software Engineer

- Developed a **vision-based perception** pipeline to detect ArUco fiducial tags using trained detection models, enabling reliable autonomous landmark recognition, target identification, and waypoint homing across unstructured terrain.
- Implemented and optimized an **iterated EKF (iEKF)** for multi-sensor localization by fusing IMU, wheel odometry, and GPS measurements, achieving robust real-time state estimation with a **30%** reduction in pose drift.
- Architected and integrated the full **MRover ROS** software stack, designing subsystem nodes, message interfaces, launch configurations, and inter-module communication pipelines for reliable end-to-end hardware operation.

Projects

EECS 467: Autonomous Robotics – University of Michigan *Dec 2025 – Present*

- Developed a real-time **2D LIDAR SLAM** system on an **MBot** using **200-particle Monte Carlo localisation** fused with wheel odometry over a 15m×15m occupancy grid map, achieving <5 cm mean localisation error.
- Implemented autonomous **frontier-based exploration** and **A* path planning** for mapless navigation across 4+ rooms, calibrating **log-odds**, **ray-casting**, and **likelihood-field sensor models** to maintain 95%+ goal rate.
- Integrated **AprilTag landmark detection** into the SLAM loop, reducing cumulative odometry drift by ∼40% compared to LIDAR-only localisation over 100m+ multi-room traversals.

Motion-Sensing Robotic Limb *Dec 2024 – Mar 2025*

- Built an **IMU-driven** finger-control interface enabling smooth, responsive, and stable multi-joint actuation.
- Tuned **sensor filtering** and feedback loops to reduce noise, improve accuracy, and refine motion calibration.
- Evaluated precision, repeatability, and controller behavior across tests to validate **mechatronic performance**.

FROST Autonomous Ice-Detection Robot *Aug 2024 – Dec 2024*

- Developed an autonomous ground robot using **IR sensing**, **thermal imaging**, and embedded control logic.
- Implemented **MBot**-based drive control and hazard-detection behavior for reliable operation on slick surfaces.
- Designed robust **CAD assemblies** optimized for precise sensor placement, stiffness, and field reliability.

AgriRobotics Systems (AutoFarm + HighWay Go) *Independent Projects*

- Designed and built autonomous agritech robots for ploughing, sowing, watering, and indoor crop growth acceleration.
- Developed a **solar-powered robotic system** to irrigate highway median plants using a robotic arm.
- Integrated embedded controllers, sensors, and actuators for reliable autonomous operation under power constraints.
- Implemented task sequencing and automation logic for coordinated motion, watering, and crop interaction.
- Designed mechanical structures optimized for stability, reachability, and efficient plant coverage.
- Evaluated system performance through real-world testing focused on energy efficiency and deployment scalability.

Skills

Languages & ML Frameworks: C/C++, Python, Julia, MATLAB, Bash, PyTorch, NumPy, OpenCV.
Robotics Software & Middleware: ROS/ROS 2, robot software architecture, nodes, launch files, tf, message passing.
Perception & CV: Multi-view geometry, camera models, calibration, feature extraction, visual odometry.
State Estimation & SLAM: EKF/UKF, particle filters, factor graphs, occupancy grids, scan matching, sensor fusion.
Planning & Auton: Graph-based planning, search algorithms, trajectory optimization, navigation stacks.
Dynamics & Control: Rigid-body dynamics, state-space modeling, PID, LQR/LQI, feedback linearization.
Simulation & Robotics Platforms: Isaac Sim, Gazebo, RViz, MATLAB/Simulink, physics simulation.
Embedded & Hardware Systems: STM32, Arduino, PWM, I^2C, SPI, encoders, IMUs, LIDAR, motor control.
Math & Algorithms: Linear algebra, probability, optimization, numerical methods, graph algorithms.

Coursework

Robotics & Autonomy:
ROB 530 – Robot SLAM(Simulataneous Mapping and Navigation); EECS 504 – Foundations of Computer Vision; ROB 430 – Deep Learning for Robot Perception and Manipulation; EECS 467/ROB 330 – Autonomous Robotics; ROB 320 – Robot Operating System/Robot Planning; ROB 415 – Controls;
Math & Dynamics:
ROB 201 – Robot Differential Equations; MATH 214 – Linear Algebra; MECHENG 240 – Dynamics;
Systems & Computing:
EECS 281 – Data Structures and Algorithms; EECS 215 – Introduction to Electronic Circuits.

Research & Instructional Experience

Undergraduate Researcher — FRoG Lab, University of Michigan
Faculty Advisor: Prof. Katie Skinner

Undergraduate Researcher — SAM Lab, University of Michigan
Faculty Advisor: Prof. Steven Ceron

Instructional Aide — ROB 201: Introduction to Differential Equations in Robotics
Instructor: Prof. Jessy Grizzle

Instructional Aide — EECS 467: Autonomous Robotics
Instructor: Prof. Xiaoxiao Du